

Predictors of surgical outcome and pathologic considerations in focal cortical dysplasia



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ABSTRACT

Background: Although surgical resection has been an important alternative treatment for patients with intractable epilepsy related to focal cortical dysplasia (FCD), the prognostic relevance of the degree of pathologic severity is controversial and there has been only limited information regarding the prognostic factors involved in the surgical treatment of refractory epilepsy in patients with FCD.

Methods: We undertook the present study to assess whether the pathologic subtypes of FCD affect surgical outcomes in patients with drug-resistant epilepsy. We also studied the prognostic roles of clinical factors and various diagnostic modalities in the surgical treatment.

Results: A total of 166 consecutive patients were included. By univariate analysis, incomplete resection of epileptogenic area ($p < 0.001$), mild pathologic features ($p = 0.01$), and the presence of secondary tonic clonic seizures (2GTCS) ($p = 0.05$) were associated with poor surgical outcomes. There was a strong tendency for patients with severe pathologic features to have MRI abnormalities ($p < 0.001$). Incomplete resection of epileptogenic area ($p < 0.001$) and mild pathologic features ($p = 0.02$) were poor independent outcome predictors on multivariate analysis. The results of MRI, scalp EEG, fluorodeoxyglucose-PET, and ictal SPECT were not associated with surgical outcomes.

Conclusions: Our study shows that there is a strong tendency for patients with severe pathologic features to have MRI abnormalities, and patients with incomplete resection, mild pathologic features, or the presence of secondary tonic clonic seizures have a high chance of a poorer surgical outcome. **Neurology® 2009;72:211-216**

GLOSSARY

2GTCS = secondary tonic clonic seizures; **FC** = febrile convulsion; **FCD** = focal cortical dysplasia; **FDG** = fluorodeoxyglucose; **FDG-PET** = fluorodeoxyglucose-positron emission tomography; **FLE** = frontal lobe epilepsy; **IED** = interictal epileptiform discharge; **mMCD** = mild malformations of cortical development; **nTLE** = neocortical temporal lobe epilepsy; **OLE** = occipital lobe epilepsy; **PLe** = parietal lobe epilepsy; **SPECT** = single-photon emission computed tomography.

Focal cortical dysplasia (FCD) is the most commonly encountered developmental malformation that causes refractory epilepsy. The pathologic features of FCD include mild disturbances, such as neuronal heterotopias or dyslamination, and severe abnormalities such as large, bizarre pyramidal neurons within the cortex and in the white matter, often associated with large balloon cells.¹⁻³ With advances in neuroimaging techniques, in particular MRI, recent studies have revealed a higher prevalence of FCD than previously estimated and have improved the preoperative identification and classification of these abnormalities.⁴

Surgical resection has been an important alternative treatment for patients with intractable epilepsy related to FCD. Although it is very important to determine the prognostic factors in identifying ideal candidates for surgery and in predicting the prognoses of individual patients, there has been only limited information regarding the prognostic factors involved in the surgical treatment of refractory FCD. The prognostic relevance of the degree of pathologic severity

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is controversial, because the presence of severe pathologic features has been associated with favorable surgical outcomes in some reports,⁵⁻⁷ whereas others have documented the opposite results.^{8,9} Moreover, few studies have analyzed the diagnostic and prognostic roles of modern presurgical evaluation techniques, such as MRI, fluorodeoxyglucose (FDG)-PET, and ictal single-photon emission computed tomography (SPECT), and their relationships to the severity of the pathology have not been adequately evaluated.

In this study, we hypothesized that the degree of pathologic severity would be related to the sensitivity of diagnostic modalities and completeness of resection, and affect surgical outcomes of refractory epilepsy in patients with FCD. Therefore, we investigated the surgical outcomes in patients with FCD and evaluated the prognostic implications of the degree of pathologic severity, clinical factors, and the results of presurgical diagnostic modalities.

METHODS Patients. Our study included 166 consecutive patients who had undergone surgical treatment for refractory epilepsy and had a pathologic diagnosis of FCD at the Seoul National University Hospital between November 1995 and June 2005. We excluded three patients with insufficient postoperative follow-up (less than 2 years), and 10 patients with other potentially epileptogenic lesions, such as tumors, vascular malformations, ischemic changes, and sequelae of trauma. We excluded 10 patients with malformations of cortical development other than FCD including periventricular heterotopia (2), polymicrogyria (7), and lissencephaly (1 patient). We also excluded 40 patients who had radiologic evidence of hippocampal sclerosis (HS) in addition to FCD, because it is possible that HS as well as FCD could act as epileptogenic areas in these patients. All patients had intractable epilepsy, despite taking the appropriate anticonvulsant drugs. We included only patients with focal resection and excluded patients with functional hemispherectomy or corpus callosotomy. Surgical outcome was divided into seizure-free and not seizure-free.

Magnetic resonance imaging. All patients underwent brain MRI. Standard MRI was performed on either a 1.0-T or a 1.5-T unit (Signa Advantage; General Electric Medical Systems, Milwaukee, WI), with conventional spin-echo T1-weighted sagittal and T2-weighted axial and coronal sequences in all patients. The section thickness and the conventional image gaps were 5 mm and 1 mm. T1-weighted three-dimensional magnetization-prepared rapid acquisition of gradient-echo sequences with 1.5 mm thick sections of the whole brain and T2-weighted fluid-attenuated inversion recovery images of 3 mm thick sections were also obtained in the oblique coronal plane of the temporal lobe. The angle of oblique coronal imaging was perpendicular to the long axis of the hippocampus. Spatial resolution was approximately 1.0 × 1.0 mm (matrix, 256 × 256 mm; field of view, 25 cm).

Functional neuroimaging. PET was performed in 151 patients during the interictal period (no seizures for more than 24

hours). Axial raw data were obtained using a PET scanner (ECAT EXACT 47; Siemens CTI, Knoxville, TN) 60 minutes after the IV injection of ¹⁸F-fluorodeoxyglucose (FDG, 370 MBq). Spatial resolution was 6.1 mm × 6.1 mm × 4.3 mm. FDG-PET images were assessed visually and by statistical parametric mapping analysis, as described previously.¹⁰ Ictal SPECT was performed in 121 patients during video-EEG monitoring. Technetium-99m (^{99m}Tc) was mixed with hexamethylpropyleneamine oxime (925 MBq) and injected as soon as a seizure started. Brain SPECT images were acquired within 2 hours of the injection. A triple-head rotating gamma camera (Prism 3000; Picker, Cleveland, OH) equipped with a high-resolution fan beam collimator was used. Interictal SPECT was also performed to identify perfusion changes. Side-by-side visual analysis of interictal and ictal images was performed and the subtraction method was used, as previously described.¹⁰ The results of FDG-PET and SPECT were defined as localizing if the predominant hyperperfusion area or the predominant hypometabolic zone was confined to the resected lobe, and diffuse or multifocal abnormalities beyond the epileptogenic area were not considered as localizing even when they are within the epileptogenic hemisphere including the epileptogenic area.

Video-EEG monitoring. Interictal and ictal scalp EEGs were recorded in all patients using a video-EEG monitoring system, with electrodes placed according to the international 10–20 system, with additional anterior temporal electrodes. In 143 patients for whom other methods gave inconclusive or discordant results, we used a combination of grids and strips for intracranial EEG. Grid and strip placements were determined by the results of presurgical evaluations. At least three habitual seizures were recorded during scalp and intracranial EEG monitoring. When necessary, preoperative and intraoperative functional mapping and intraoperative electrocorticography were also performed.

A localizing pattern of ictal-onset rhythm/interictal spike was defined as a localized ictal rhythm/interictal spike in the electrodes of an epileptogenic lobe or two adjacent electrodes. In patients with multiple areas of ictal-onset rhythm/interictal spike, we defined as localizing only when the whole areas are finally considered epileptogenic, and did not include when the abnormalities were multilobar within the epileptogenic hemisphere including the epileptogenic area.

Surgery and pathology. The surgical area was decided based on the clinical, neuroimaging, and electrophysiologic results. In our center, the resection margin for epilepsy of neocortical origin was defined by 1) the presence of either a discrete lesion on MRI with compatible ictal EEG or a massive and exclusive ictal-onset zone confirmed by intracranial EEG and 2) the absence of eloquent cortex. Complete resection was defined by resection of all of the MRI-visible lesions or areas of ictal onset, persistent pathologic delta slowing, >1/second frequent spikes, and intermittent gamma wave by intracranial EEG.

The diagnosis and classification of pathologic cortical dysplasia were according to the system of Palmini and colleagues: ectopically placed neurons only (mMCD), isolated architectural abnormalities (FCD 1A), additional immature or giant neurons (FCD 1B), presence of dysmorphic neurons (FCD 2A), and additional balloon cells (FCD 2B).¹

Statistical tests. The association between the pathologic severity of FCD and the diagnostic sensitivity of various diagnostic modalities was tested with a linear association χ^2 test. We compared the clinical characteristics of the seizure-free group and the non-seizure-free group with Student *t* test or the Mann-

Whitney *U* test for continuous variables and with a χ^2 test for categorical variables.

Variables with a *p* value of <0.2 on univariate analysis were included in a multiple logistic regression model, to identify the predictors of surgical outcome. In multiple logistic regression, backward elimination was used to select variables to be maintained in the final model, using a *p* value of <0.05 as the criterion for the significant associations. All analyses were conducted using SPSS version 12.0 (SPSS Inc., Chicago, IL) and STATA version 9.2 (STATA Corp., College Station, TX). A *p* value <0.05 was considered significant.

RESULTS Patients and surgical outcomes. In our surgical cohort, 956 patients underwent surgical treatment for refractory epilepsy excluding vagus nerve stimulator implantation, and the pathologic diagnosis of FCD was made in 206 (21.5%) patients. Forty patients were excluded for the radiologic evidence of HS, and 166 (17.4%) patients were finally included for the present analysis.

The subjects consisted of 102 men and 64 women, and the mean age of the patients at the time of surgery was 24.7 ± 10.2 years (range, 3–51 years). The mean age at seizure onset was 11.5 ± 7.0 years (range, 0.5–30 years) and the mean interval between seizure onset and surgery was 12.8 ± 7.8 years (range, 0.5–37 years). A history of febrile convulsions was present in 18.1% (30/166) of patients, and 74.7% (124/166) of patients experienced secondary generalized tonic clonic seizures (2GTCS).

The mean length of postoperative follow-up was 7.94 years. The proportion of seizure-free patients was 56.6% (94/166) during the follow-up for more than 2 years, whereas 43.4% (72/166) of patients did not become seizure-free. Changes of outcome during the follow-up period were observed in 17 of 166 patients. The running-down phenomenon was observed in 11 of 94 patients who became finally seizure-free, and reappearance of seizures after initially seizure-free state occurred in 6 of 72 patients who were not seizure-free.

Postoperative mortality and morbidity. No postoperative mortality was found. Postoperative complications occurred in 15 (9.0%) patients. Postoperative infections occurred in seven patients including four meningitis, one skull osteomyelitis, and two scalp wound infections. Four incidents of postoperative hematoma were noted after intracranial electrode placement, which required emergency decompression. A postoperative neurologic deficit was noted in four patients, but all the neurologic deficits resolved spontaneously during the follow-up period.

Surgical outcomes according to different pathologic subtypes of FCD. Pathologically, 21 patients showed characteristics of mMCD, 89 patients of FCD 1A, 28 patients of FCD 1B, 12 patients of FCD 2A, and

16 of FCD 2B. There was a strong tendency toward better surgical outcomes at the last follow-up visit in patients with more severe abnormalities, because only 42.9% of patients with mMCD, 51.7% of those with FCD 1A, and 60.7% of those with FCD 1B became seizure-free, compared with 66.7% of patients with FCD 2A and 87.5% of those with FCD 2B who became seizure-free (*p* = 0.003).

Surgical outcomes according to completeness of resection. Complete resection of potentially epileptogenic area was performed in 111/166 (66.9%) patients, and the completeness of resection and pathologic severity were not associated, as shown in table 1. However, the completeness of resection was an important prognostic factor on univariate (table 2) and multivariate analysis (table 3). We could not perform complete resection in 55 patients because the epileptogenic zone included portions of eloquent area in 26 patients, and the epileptogenic area was often multifocal (10 patients) and widespread (19 patients).

Diagnostic sensitivities of various modalities and their relationships to pathologic subtypes. MRI detected focal abnormalities in the resected area in 42.2% of patients. Interictal scalp EEG showed unifocal epileptiform discharges to the resected lobe in 42.7% of patients, and ictal scalp EEG correctly localized them to the resected lobe in 76.5% of patients. FDG–PET showed concordant focal hypometabolism in 71.4% of patients and ictal SPECT showed concordant focal hyperperfusion in 60.1% of patients.

The diagnostic sensitivities of various diagnostic modalities in relation to pathologic classification are summarized in table 1. There was a strong tendency to have identifiable lesions on MRI (*p* <0.001) in patients with severe pathologic features. The diagnostic sensitivities of interictal/ictal EEG findings, FDG–PET, and ictal SPECT were not related to the severity of the pathologic features.

Prognostic factors for seizure-free outcomes. On univariate analysis, the presence of 2GTCS (*p* = 0.05) was associated with poor surgical outcomes. The patients were divided into two subgroups: the mild pathologic group (mMCD, FCD 1A, and FCD 1B) and the severe pathologic group (FCD 2A and 2B). Patients in the severe pathologic group had a higher chance of becoming seizure-free than did patients in the mild pathologic group (*p* = 0.01; table 2).

Because there was a strong tendency for patients with severe pathologic features to have focal lesions on MRI, the prognostic role of pathology may depend on the results of MRI. However, the severe pathologic features were related to a high rate of becoming seizure-free, even after adjustment for the effect of MRI (*p* = 0.04).

Table 1 Completeness of resection and diagnostic sensitivity according to pathologic subtypes

	mMCD	FCD 1A	FCD 1B	FCD 2A	FCD 2B	p Value
Completeness of resection of potentially epileptogenic zone	13/21 (61.9)	57/89 (64.0)	20/28 (71.4)	7/12 (58.3)	14/16 (87.5)	0.131
MRI						
Diagnostic sensitivity	19.0 (4/21)	30.3 (27/89)	60.7 (17/28)	75.0 (9/12)	81.3 (13/16)	<0.001
False negative rate, %	81.0	69.7	39.3	25.0	18.7	
Interictal EEG						
Diagnostic sensitivity	61.9 (13/21)	40.4 (36/89)	50.0 (14/28)	16.7 (2/12)	37.5 (6/16)	0.206
False negative rate, %	38.1	59.6	50.0	83.3	62.5	
Ictal scalp EEG						
Diagnostic sensitivity	76.2 (16/21)	79.8 (71/89)	75.0 (21/28)	66.7 (8/12)	68.8 (11/16)	0.352
False negative rate, %	23.8	20.2	25.0	33.3	31.2	
FDG-PET (151)						
Diagnostic sensitivity	66.7 (12/18)	66.3 (55/83)	78.3 (18/23)	41.7 (5/12)	93.3 (14/15)	0.269
False negative rate, %	33.3	33.7	21.7	58.3	6.7	
Ictal SPECT (121)						
Diagnostic sensitivity	80.0 (12/15)	56.3 (36/64)	40.9 (9/22)	71.4 (5/7)	76.9 (10/13)	
False negative rate, %	20.0	43.7	59.1	28.6	23.1	
Invasive monitoring (143)						
Diagnostic sensitivity	68.4 (13/19)	85.9 (67/78)	86.4 (19/22)	80 (8/10)	92.9 (13/14)	0.189
False negative rate, %	31.6	14.1	13.6	20.0	7.1	

Values are n (%) or %.

mMCD = mild malformations of cortical development; FCD = focal cortical dysplasia; FDG-PET = fluorodeoxyglucose-positron emission tomography; SPECT = single-photon emission computed tomography.

Multivariate analysis identified incompleteness of resection ($p < 0.001$) and mild histopathologic features ($p = 0.02$) as independent significant prognostic factors for a poor surgical outcome (table 3).

DISCUSSION Consistent with our observation, complete resection of MRI-visible lesions or the epileptogenic zone has been associated with favorable surgical outcomes.^{5,9,11,12} Several studies documented the prognostic values of several clinical factors including seizures of short duration or early epilepsy surgery; we could not replicate these reports.¹²⁻¹⁴

Our study shows that patients with severe pathologic features have a higher chance of becoming seizure-free after surgery. We first assumed that the prognostic effect of pathologic severity in our study was attributable to the different sensitivity of MRI in detecting FCD, because patients with severe FCD pathology have a higher chance of presenting with abnormalities on MRI. However, the prognostic effect of pathologic severity was still strong after we statistically adjusted for the effect of MRI. The positive prognostic effect of severe pathologic features has also been documented in previous reports, which predominantly included patients with identifiable lesions on MRI.^{5-7,15} The prognostic role of pathologic severity may be related to the completeness of surgi-

cal resection. Although the epileptogenic zone in FCD does not always correspond to the area with the most severe pathologic features, it is generally accepted that the epileptogenicity of FCD is usually associated with its pathologic severity. Because the grading of pathologic severity is usually based on the features of the most severely affected area, mild pathologic grades may reflect incomplete surgical resection of the epileptogenic zone in some patients. Alternatively, differences in prognoses may be associated with the different sensitivities of diagnostic modalities other than MRI. It can be assumed that severely dysplastic areas are more epileptogenic, regardless of the presence of an MRI abnormality, and that highly epileptogenic areas have a greater chance of accurate localization by other diagnostic modalities, such as scalp EEG and functional neuroimaging.

The presence of 2GTCS was associated with poor surgical seizure-free outcomes in our study. The negative prognostic role of 2GTCS in epilepsy surgery for FCD has been observed in patients with mTLE,^{16,17} frontal lobe epilepsy,¹⁸ and FCD.^{11,12} The presence of 2GTCS may be a marker for more widespread cortical involvement and a widely distributed epileptogenic area,¹² or continuing generalized epileptic activity may lead to widespread cortical struc-

Table 2 Clinical characteristics and outcomes after surgical resection in 166 patients with epilepsy with focal cortical dysplasia: Univariate analysis of prognostic factors for seizure-free outcomes for at least 2 years after surgery

	Seizure-free (n = 94, 56.6%)	Not seizure-free (n = 72, 43.4%)	p Value
Age at operation (y, mean $\pm$ SD)	24.7 $\pm$ 10.2	23.6 $\pm$ 8.7	0.49
Age at onset (y, mean $\pm$ SD)	11.4 $\pm$ 7.5	11.5 $\pm$ 6.4	0.96
Duration of epilepsy (y, mean $\pm$ SD)	13.3 $\pm$ 8.2	12.2 $\pm$ 7.2	0.37
Men/women	56/38	46/26	0.57
Seizure frequency per month (mean $\pm$ SD)	21.4 $\pm$ 47.9	20.9 $\pm$ 55.6	0.94
Presence of 2GTCS	64	60	0.05
Presence of FC	17	13	1.00
Complete resection of epileptogenic area	77	34	<0.001
MRI, concordant focal lesion	45	25	0.09
Interictal EEG, focal IED	41	30	0.80
Ictal EEG, localized ictal onset zone	68	59	0.15
PET, concordant focal hypometabolism (n = 151)	55/82	49/69	0.60
Ictal SPECT, concordant focal hyperperfusion (n = 121)	38/71	34/50	0.11
Invasive study	84	59	0.17
Mild pathologic characteristics	72	66	0.01
Epilepsy syndrome			
FLE	30	23	0.16
nTLE	24	28	
PLE	8	8	
OLE	9	3	
Multilobar	23	10	

2GTCS = secondary generalized tonic clonic seizure; FC = febrile convulsion; IED = interictal epileptiform discharge; FLE = frontal lobe epilepsy; nTLE = neocortical temporal lobe epilepsy; PLE = parietal lobe epilepsy; OLE = occipital lobe epilepsy.

tural and functional changes, including secondary epileptogenesis in interconnected brain areas that are not removed by surgical resection.¹⁹

Whereas approximately 80% of patients became seizure-free after surgical treatment for mTLE related to HS or lesional epilepsy, such as primary brain tumor or vascular malformations,^{16,20} the efficacy of surgical treatment for FCD was consistently less favorable, with approximately 33–75% of individuals becoming seizure-free.^{5,6,12,21} Simple resection of the lesion, as performed in patients with mTLE or lesional epilepsy, may be a feasible surgical option as

long as the lesion is completely removed. However, the complete removal of FCD is often difficult because the demarcation of the lesion is frequently poor, and dysplastic tissues tend to be more extensive than is apparent on MRI.^{11,22} Furthermore, the seizures can originate in either the area of MRI abnormality or the area adjacent to the lesion, and MRI frequently does not show any abnormalities in patients with pathologically proven FCD.²³ The frequent overlap of the epileptogenic zone with eloquent cortex may also be involved in the poor surgical outcomes.²⁴ Evidence indicates that even patients with MRI abnormalities who have resective epilepsy surgery for FCD have worse surgical outcomes than those of patients who have surgery for other focal lesional epilepsy syndromes.²⁵ The seizure-free rate in our study (56.6%) was no better than those of previous reports, but this result may be significant, considering that more than half the patients in our study failed to show definable abnormalities on MRI.

Although a direct comparison of MRI and other diagnostic modalities may be difficult, it is clear that the localization of a lesion by MRI can both enhance our confidence and provide more information than do FDG–PET or ictal SPECT, especially when the localization is supported by an electrophysiologic study. MRI can also be helpful in planning appropriate intracranial electrode placement. Imaging findings may favor surgery in some patients who might otherwise be rejected for surgery.²⁶ Although only a trend toward better surgical outcomes was observed in patients with definable lesions on MRI ($p = 0.09$), we believe that the recruitment of a large number of patients may confirm the prognostic value of MRI in epilepsy surgery for FCD.

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Table 3 Clinical characteristics and outcomes after surgical resection in 166 patients with epilepsy with focal cortical dysplasia: The final model of multivariate analysis of prognostic factors for seizure-free outcome for ≥ 2 years after surgery

Factors	OR	95% CI	p Value
Mild pathologic characteristics	0.305	0.108–0.857	0.02
Complete resection of epileptogenic area	4.939	2.405–10.140	<0.001

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